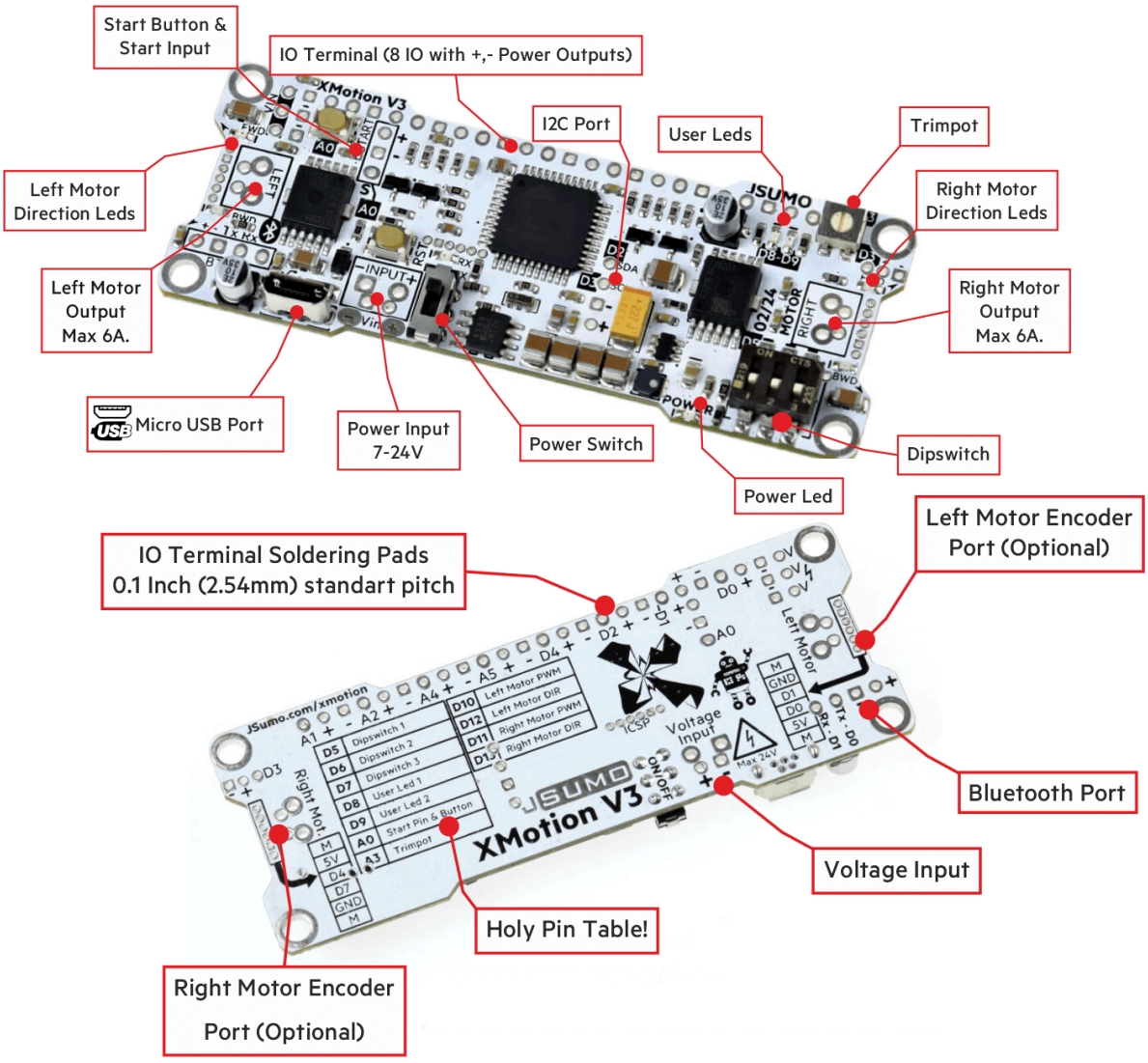
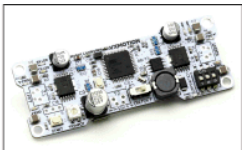
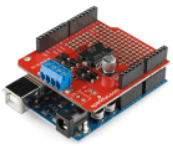
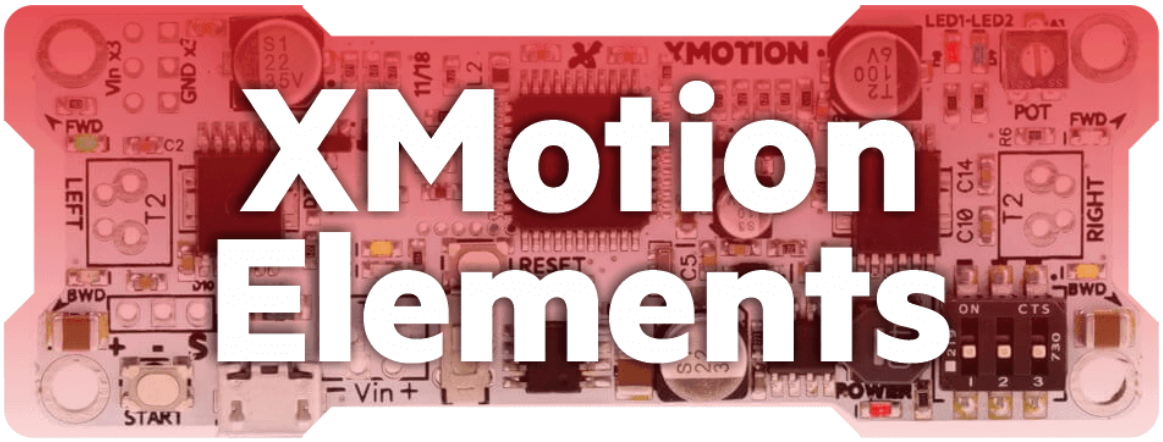


What is that strong features?

Small dimension. It was very important factor at our design, small but at same time stabilized design. We managed to make 80 x 30 x7 mm dimension board without rectricts.
6 Ampere x 2 Motor drive outputs. This outputs can supply high currents and gives juice more than enough for small robotics projects. When compared with traditional boards 6 Ampere triples, doubles the other boards. Also they are protected for short circuits.
Voltage input protected for reverse connection. At full current voltage drop at reverse connection mosfet is only 60 millivolts!
Built-in regulator can work up to 24V! (For generic Arduino boards this voltage is 15V with lots of heat) XMotion never heats.
At bottom you can program with micro USB. Main Mcu (**ATMEGA 32U4**) preloaded with Arduino Leonardo Software. Very easy to program.
Sturdy micro USB socket that can withstand many years without any break thanks to pinned strong design.
Board has analog and digital input elements like **trimpot, button & dipswitch**. You can create trimpot based speed control, or dipswitch based robot tactics & modes.
You can connect up to 9 sensors or interface circuits with thoruh hole pads of controller IO terminal.
For more understanding please look to comparison table

				
	XMotion	**Arduino Uno +Ardumoto**	**Baby Orangutan**	**SkyDragon**
Dimensions	80mm x 30 mm x 12mm	77mm x 53mm x 24mm	30mm x 18mm x 6mm	68mm x 34 mm x 8mm
Price	$49.00 (With Guarantee & Support)	$45	$19.95	$40
Board Voltage Input	7V-28V (No heat)	7V-15V (With heat)	5-13.5V (With heat)	7-15V (With heat)
Motor Drive Current	6 Ampere x 2	2 Ampere x 2	1 Ampere x 2	2 Ampere x 2
Voltage Regulator Current	1600 mA (No Heat)	100mA (Heat)	100ma (Heat)	100ma (Heat)
Built-In USB Connectivity	Yes (Micro USB)	Yes (USB B)	No	Yes (Mini USB)
Learning Curve	Easy	Very Easy	Hard	Easy
Harsh Usage (Noisy Environment, Short Circuit etc...)	Yes	No	No	No

Now, if we summarize the features of XMotion:



Microcontroller
Atmega 32U4 Arduino Leonardo Bootloader uploaded. Flexible and easy to develop for all type robotics projects.

Voltage Regulator
Switching Mode regulator for 7-24V input Voltage.
5V 600ma Output current for all sensor connections.

Motor Drivers
Modern Mosfet based 6A outputs x 2. Short circuit proof. Easy control with Dir/PWM control. (We have examples at XMotion library)

User & Motor Direction Leds
2 User Leds free to program. 4 Motor direction leds for easy recognition the direction without using motor. (2 Channel x 2 Leds)

Start Button & Pin
Compatible with all robot start modules. Paralelled with push button.

Dipswitch
3 Position, 8 combination digital switch.

Capacitors
Multilayer low ESR ceramic capacitors distributed along the board.

Serial Resistors
All MCU pins have serial connected current limiter resistors.

Trimpot
Analog input element. 0-5V Output. Can be used for changing speed, time vs.. variables. (We have example for trimpot at library examples)

Fully Protected
Reverse Battery, 5V short-circuit, motor output short circuit, high voltage, over current & over-temprature for regulator & motor driver.

Micro USB Connection
Xmotion can be easily programmed with on-board Micro USB port. Programming cable included with every Xmotion board!

Power Terminals
Motor and battery terminal comes as unsoldered for flexibility. This pads accepts 3.5mm screw terminals, 2.54mm male headers or direct soldering.

I/O Terminals

9 input, output through hole terminal at bottom side of board. Pins can be used as 5 digital & 4 analog or all pins can be used as 9 digital I/O.

Voltage Protected I/O

8 of 9 inputs are voltage protected up to 15 volts. It means that you can connect directly 12V Industrial sensors to inputs without any divider resistors! (Yes, you can connect all Omron, Pepperl Fuchs and other brand 12V working PNP output sensors)

Bluetooth & I2C Expansion Ports (New at V3)

You can connect Bluetooth modules (such as HC-06) and I2C modules (such as MPU6050 IMU Module or OLED display modules) via the expansion ports on the board easily.

Encoder Motor Connection Ports (New at V3)

The small hole sockets (1.27mm pitch x 6) in front of the motor outputs on the XMotion V3 board are connections optimized for MCU interrupt pins for direct connection of encoder motors. You can directly assemble encoder motors that you can use in projects such as the Maze Solving robots.

Weight: 13 grams. **Dimensions:** 80mm x 30 mm x 7mm.

WHICH **SENSORS** ARE SUITABLE?

All digital and analog output sensors can be used with XMotion.
With 8 protected Inputs also you can connect directly PNP output 12V sensors like Omron, Keyence sensors... Here some models suitable to XMotion

JS40F Digital Opponent Sensor
MZ80 Infrared Sensor
Omron E3Z-D62-82 (No resistor divider needed!)
Pepperl Fuchs MLT1000 (No resistor divider needed!)
Banner Q20NDXL
Parallax Ping Sensor
HCSR04 Ultrasonic Sensor
QTR1A and QTR1RC Line Sensors

NEXT PAGES FOR XMOTION

Please visit this page: https://www.jsumo.com/xmotion
Read the XMotion Basics page: https://blog.jsumo.com/xmotion-basics-xmotion-101/
How to Make mini sumo robot with XMotion: https://blog.jsumo.com/how-to-make-mini-sumo-robot-with-xmotion/

20+1 Reasons for Choosing XMotion

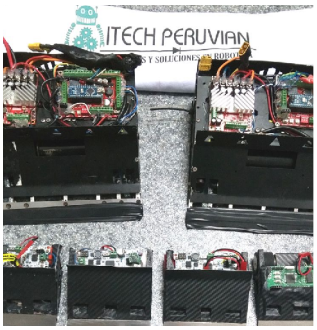
1. Broad working voltage (7V to 24V)
2. High Current Motor Drive Outputs (6Ax2)
3. Switching mode Power supply
4. Ready made XMotion Library
5. Reverse Battery Protection.
6. Motor output short circuit protection
7. 5V Regulator output short circut protection.
8. Small dimension (80x30x7mm).
9. Lightweight (13 grams).
10. Programmable user leds (2)
11. 8+1, 9 IO for sensors and extra modules.
12. Micro USB programming
13. 600mA Regulator output.
14. Easy sensor connection with + & - pads.
15. Motor direction leds.
16. Electrical noise-free design.
17. 3 Position dipswitch input.
18. Built-in trimpot for easy setting variables.
19. Current limiter group resistors.
20. Arduino IDE, no extra driver installation

21. Guaranteed, 100% tested controller board for your success.

Release Logs

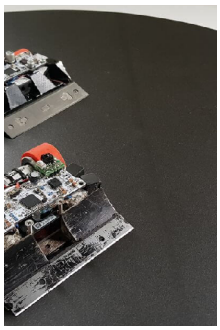
XMotion V.3 02/24 - Moving to 4 layer design, vertical Micro USB update, IO pads changed to through hole from SMD, Added Encoder Motor Ports, Bluetooth and I2C ports, bulk MLCC capacitors added for better transient response, removed Voltage divider sensor and total free I/O increased to 9 from 8.
Motion V.2 12/22 - Micro USB update, Regulator update, returning to original drivers.
XMotion V.2C - Corona Update, Motor driver change due to finished original motor driver stocks.
Xmotion V.2 - Regulator update & shematic update for making more rugged, that allows higher voltage Inputs up to 15V.
XMotion V.1.2 - Board layout update
XMotion V.1.1 - Micro USB layout updated
XMotion V.1 - Launch

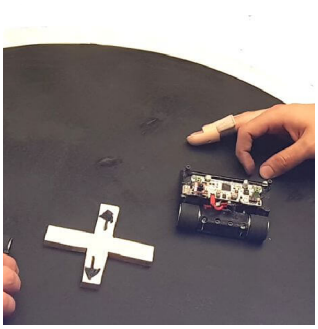
XMotion Controller based robots from worldwide customers












Tags